Dual-ligand screen / Setup

48 pairs, then 22 pairs + L17 reference. One 48 x 2 mL Para-Dox, reused with **71 fresh reaction vials**. Manual screw caps; no stir bars. Keep reaction and LC maps separate.

Before the clock

- Fit **p300 single GEN2 left; p300 multi GEN2 right**. Single only for 13 mm-pitch reaction vials. Check block/latch, offsets, cap height <=45 mm and tips >1 mm above bottoms. Simulate released Python locally.
- Use matching `dosing.csv`, `stock_preparation.csv`, `deck_loads.csv`, `lc_sequence.csv` and `round2_predose.csv`. A software colleague handles release and LC imports. Stocks and initial scoring are outside the clock.
- Have 71 reaction vials/caps, **two final 96-well LC plates** matching the VWR 2.2 mL geometry, compatible pierceable seals and fresh tips; standards separately. Confirm plate fit on shaker AND autosampler, needle depth and **full 1.7-min LC cycle**.

Prepare stocks

Stock	Bulk preparation / use
SM in DCM	**500 mM, 10 mL:** 805.8 mg SM at 100% assay; capped 20 mL glass vial. Dose 50 µL; evaporate fully.
Ligands / Pd	**40 mM in toluene.** Use CSV masses/reserves. Open and prepare ligands in glovebox. Keep reserves capped.
Mesitylene	**100 mM in toluene, 5 mL:** 60.095 mg. Dose 25 µL AFTER DCM removal.
Quench	**100 mL, 9:1 ACN/water +0.15% formic acid:** 150 µL acid, make to volume. No IS.
LC diluent	**125 mL, 1:1 water/ACN +0.1% formic acid:** 125 µL acid, make to volume. No IS.

Correct masses for assay. **Load stage-specific CSV volumes, not bulk preparations.** Refill R2 from reserves; ordinary plastic source racks are sufficient.

Deck

Slot	Contents
1 / 4	Ligand source racks; exact CSV addresses; L07 absent
2	A1 SM/DCM; A2 Pd; A3 toluene; A8 mesitylene. Quench A4/A6/A9; diluent A5/A7/A10/A11/A12
3 / 10	Reaction block: staff move it by hand at App prompts; pipetting / heater-shaker
6	Final 96-well 2.2 mL LC plate: fill here; move to 10, then directly to HPLC
5 / 8 / 9	Single tips / multi tips / multi tips. Keep 7 and 11 empty

Calibration during hold one - 14 injections

Each SM and product: blank + **10, 50, 150, 350, 800 µM**, plus independent **666.7 µM** check. **Mesitylene stays at 66.7 µM in every standard.** Use matched matrix and calibration recipes. HTEL exports UV areas; Python calibrates and calculates yields. The software colleague handles method/RT settings and pilot checks.

Dual-ligand screen / Run

Operator / start: _____ LC method / channel: _____

Follow App pauses. Hand-tighten reaction-vial caps before heating/shaking; the LC plate shakes uncovered. Record actual times and deviations.

Step	Action
1 / SM	Dose **50 µL SM/DCM** into 48 vials. At pause, move off deck; evaporate fully by HTEL's ventilated procedure. Cool; return to slot 3.
2 / Reaction	Add **25 µL mesitylene +37.5 µL ligand A +37.5 µL ligand B +75 µL Pd +75 µL toluene =250 µL**. Hand-cap; move/latch as prompted. **95 °C, 500 rpm, 4 h.**
3 / During hold	Run calibration. Hand-dose **23 labelled NEW vials with 50 µL SM/DCM each** in an off-deck rack. Evaporate fully; cap dry SM. No mesitylene/catalyst yet.
4 / Cool	Move the sealed block off the heater. Cool vials before opening; App also waits for module <=30 °C. Load workup sources.
5 / Quench	Add **500 µL quench** per vial. Hand-recap ALL. Shake **1500 rpm, 2 min**, heater off; return to slot 3.
6 / Sample	Let bubbles settle after shaking; uncap the indicated reaction vials. Robot adds **20 µL extract to 980 µL diluent** with fresh sample tips.
7 / LC handoff	Recap reaction vials. Move the completed LC plate **6** → **cooled shaker 10** at App prompt. Robot shakes **UNCOVERED, heater OFF, 1000 rpm, 2 min** (locally confirmed splash-free speed). Move **10** → **HPLC**, apply compatible seal. Inject directly from this plate; **no manual dilution/mixing or vial transfers**.
8 / R2 setup/select	During R1 LC, archive sampled R1 vials; fit the 23 dry predosed vials into the SAME block and check orientation. After all 48 results pass QC, software selects 22 new pairs +L17 reference. Add catalysts after selection.
9 / R2	Refill R2 deck loads; **uncap all 23 NEW vials before dosing. No SM redose.** Add stocks; reference uses **75 µL L17**, no B. Cap; repeat 4 h reaction/workup. One new LC plate; **H3 is diluent only: do not inject.**

Stop the clock after **23 final injections and quantitative analysis**. Save exports, maps, QC, protocols, times and robot JSON logs.

If a transfer looks wrong

Pause for layers, solids at aspiration, persistent bubbles, air uptake or tip contact. Repeated aspiration does not fix solubility. Reuse tips only for approved same-stock/noncontact dosing; samples always get fresh tips. Replace an uncertain dilution with a fresh documented one; do not guess a top-up.

12-hour operating target: 8 h reaction holds +120.7 min LC; **119.3 min for all other tasks and contingency**. Calibration/R2 SM predosing overlap hold one; R2 block changeover can overlap R1 LC after sampling. Confirm actual cycle/handling times before release. **Four LC-plate moves; no R1 split.** Full procedure and local commands.